UNIVERSITÀ DEGLI STUDI DI BERGAMO

SUCCESSFUL BIBLIOGRAPHIC RESEARCH

A practical guide to find and use academic resources

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Library services
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Today’s agenda

- **Bibliographic research: workflow and search strategies**
- **PRISMA for systematic literary review**
- **Finding and accessing resources through the library**
- **Focus on: Scopus**



What is bibliographic research?

Collection and critical analysis of publications —such as articles, books, conference proceedings, and theses— ***to gather and evaluate existing information published in the literature on a specific topic.***

Academic literature = peer-reviewed scientific publications

MAIN GOALS

- To give an **overview of existing knowledge** and **research trends**
- To **support arguments** and provide evidence in academic writing
- To **identify gaps, controversies, and open questions** in current literature
- To **reduce the risk of duplication** of previous work and ensure originality



FINDING COLLECTION EVALUATION

The steps of bibliographic research

Formulate a research question

Identify relevant keywords

Search academic databases and other reliable resources

Refine the results

Review abstracts to assess the relevance of each publication

Access the full-texts

Save and organize citations



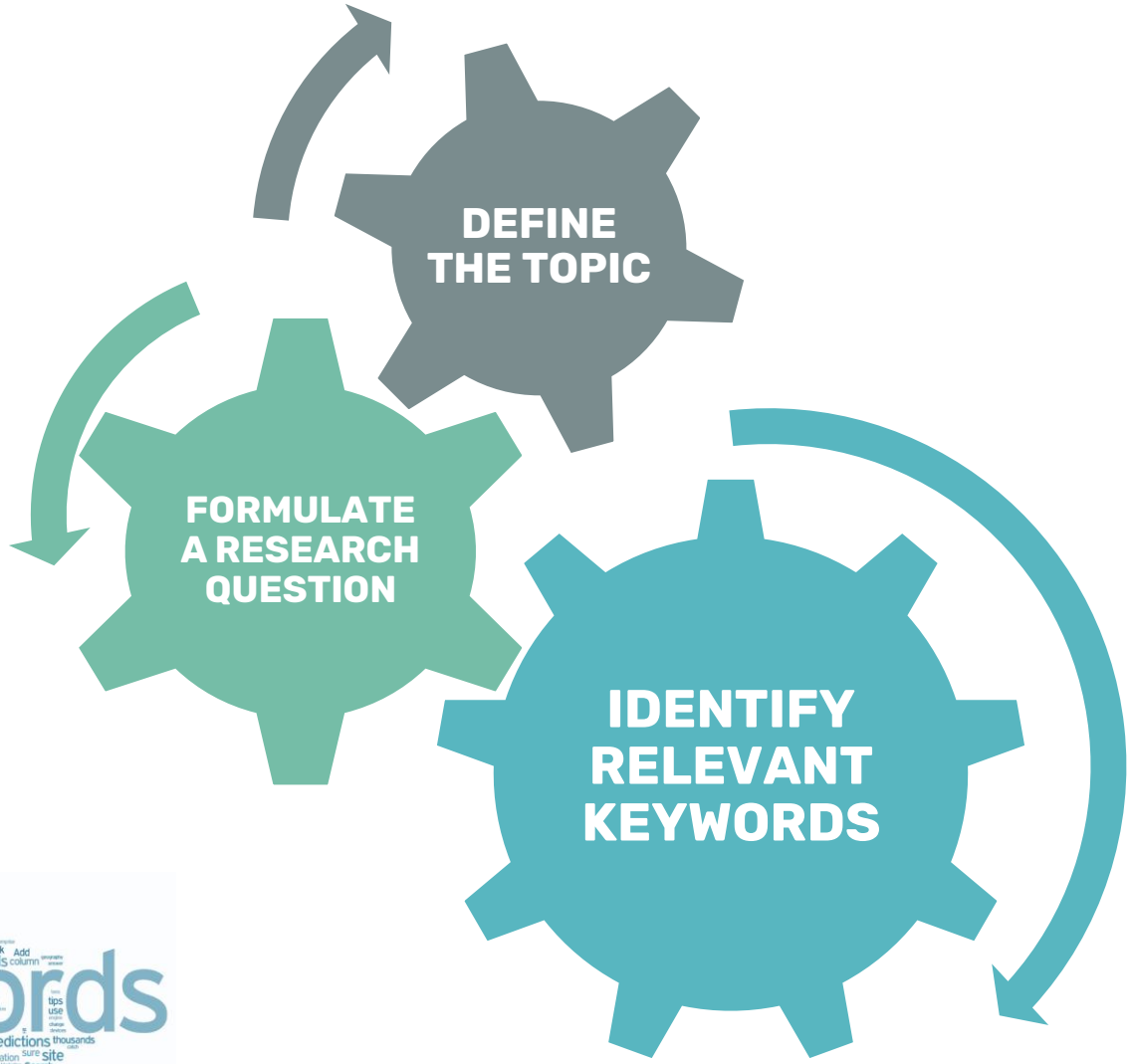
Getting started

- Clearly define the **topic and scope** of your research project
- Formulate a **precise research question** to guide your investigation
- Identify a set of **relevant keywords** and phrases to use for searching **in bibliographic database**



The more specific your keywords are, the more focused and relevant your search results will be, minimizing the “noise” (out-of-topic publications)





Exploration and exploitation

Begin with an exploratory search to identify a few **relevant articles**

Review the references to quickly discover earlier publications

Check the **citations** to **track the evolution** of the research topic

This approach helps **map the development** and current landscape of your research area



How to build a search query

Introduction to Boolean Operators for building complex queries

Logical connectors used to combine or exclude keywords in database searches

Help to **refine search results** to be more relevant and precise

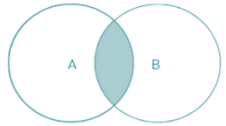
Save time and increases the quality of your research

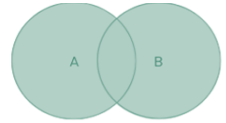


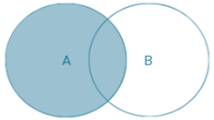
*Mastering these operators can greatly improve search precision and efficiency and help balancing between **reducing irrelevant results** and **avoid excluding something important***

How to use Boolean operators

AND	•Search for documents containing **all the keywords to narrow results** •Example: *inflation AND unemployment* → find studies discussing both inflation and unemployment
OR	•Search for documents containing **either keyword to broaden results** •Example: *GDP OR gross domestic product* → find studies using either terms
NOT	•**Exclude documents** containing a term to refine results •Example: *investment NOT real estate* → find studies on investment excluding real estate
Quotes " "	•Search for **exact phrase** •Example: *"monetary policy"* → find documents where the phrase appears exactly
Truncation *	•Use an asterisk to **find variations of a root of a word** •Example: *econom** → find economy, economic, economics, etc.
Parentheses ( )	•**Group terms** to structure complex searches •Example: *(inflation OR deflation) AND GDP*
Combine BO	•Built more **complex search query** •Example: ("Artificial intelligence" OR AI) AND "supply chain" NOT healthcare







Evaluating the sources

Primary sources: provide original data, results or evidence, such as empirical studies, original research, official documents, datasets

Secondary sources: provide interpretations, reviews and analyses of the primary sources

Tertiary sources: provide summaries, such as encyclopedias, textbooks





This hierarchy helps you **to prioritize the most relevant and original** sources

PRISMA for systematic literary review

PRISMA

What is it?

- **P**referred **R**eporting **I**tems for **S**ystematic Reviews and **M**eta-**A**nalyses
- It is a reporting **guideline framework** that includes a **flow diagram** and a **checklist** for systematic literature reviews
- First published in 2009, updated in 2021 through PRISMA 2020 Statement paper
- Initially developed in the health sciences, now used in different disciplines
- It is a **framework for reporting** how systematic review was planned and carried out

It tracks and reports sources screening and evaluation process

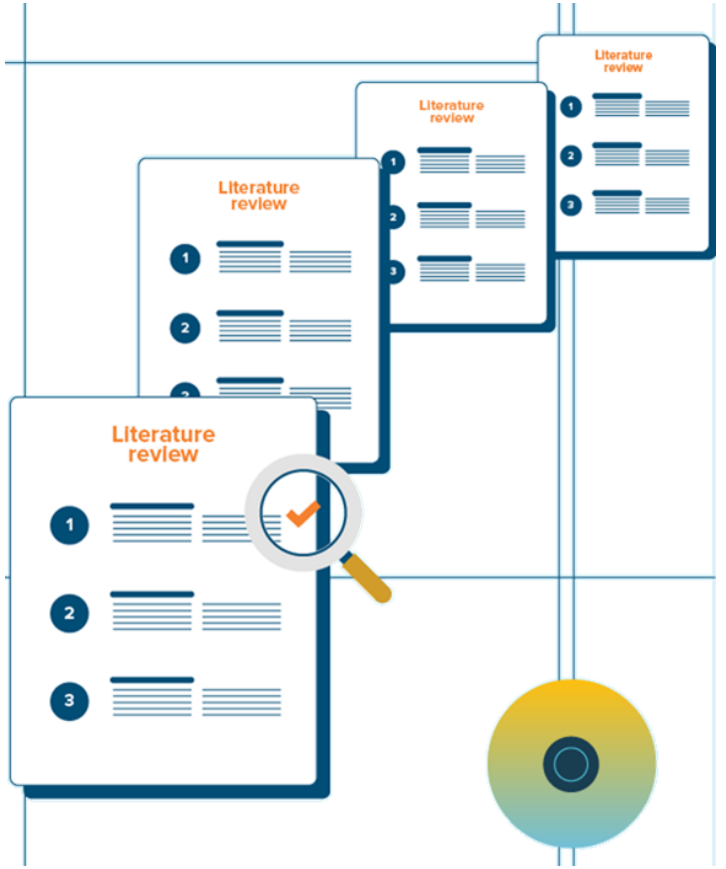
Why use it?

- Helps to **define the aim** of the review for a **comprehensive bibliographic research**
- **Track the criteria** for including or excluding the results
- Report the literary review process transparently
- Enhances **reproducibility** in literature reviews
- **Tracks the sources screening and evaluation process**

PRISMA overview

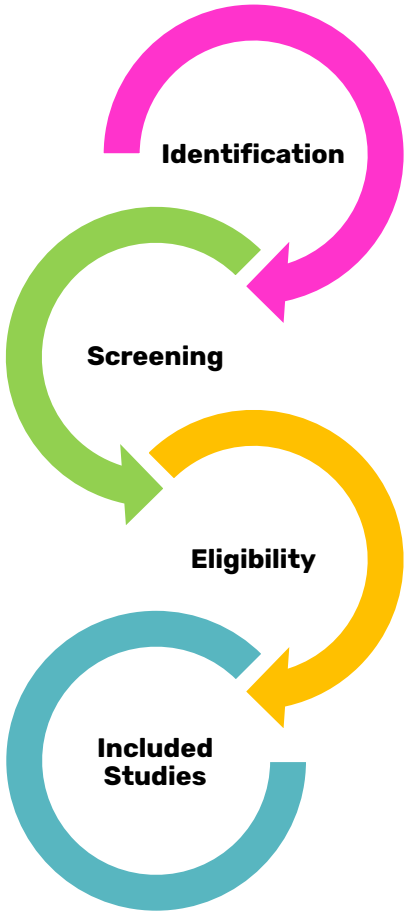
Flow diagram: **visually summarizes the screening process**, recording the number of articles found and then makes the selection process transparent by reporting on decisions made at various stages

Checklist: a list of **27 items that should be included in the final** systematic review, covering the title, abstract, introduction, methods, results, and discussion



PRISMA flow diagram

Track the sources **evaluation** and **screening** process

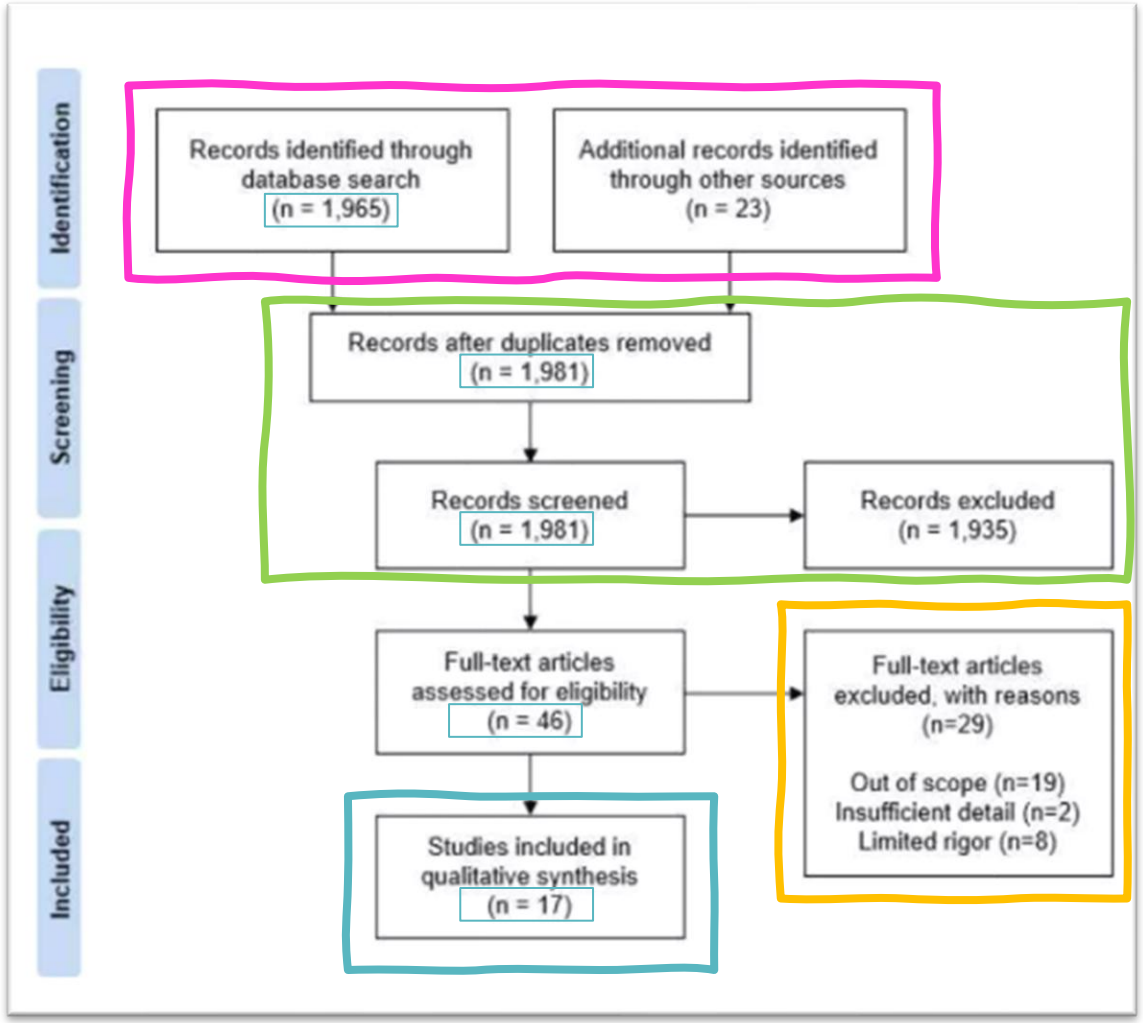


Annotate the total number of records you found after searching different databases

Track the removed duplicates and the excluded records (by title, abstract, language, YOP…)

Evaluate the full-text and report the reason why you chose to remove some articles

Record the final number of the publications included



PRISMA check-list

- ✓ Use it as **step-by-step guide** for writing your review
- ✓ It ensure **no important element is omitted**
- ✓ It covers every part of the review report (title, abstract, introduction, methods, results, discussion) and organize all the relevant information that must be included in a systematic literary review

PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Title
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Abstract
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Introduction
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Introduction, Results
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Methods, Appendix1
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Methods, Appendix1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Methods, Appendix2
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Methods, Appendix3

PRISMA takeaways

It includes a **flow diagram** that **visually maps** study selection showing how many records were **collected**, **screened, excluded, and included**

It uses a **27-item checklist** to ensure **complete** and **standardized reporting** of the systematic review process

It enhances **transparency, efficiency and reproducibility** of the systematic review process



AI for literary review

AI can support

- **Turn a topic into** searchable **concepts**
- **Expand keywords**, synonyms and controlled vocabulary
- Help **refine Boolean search** strings
- Assist with **deduplication and screening** step
- **Summarise** full texts or extract data for checking



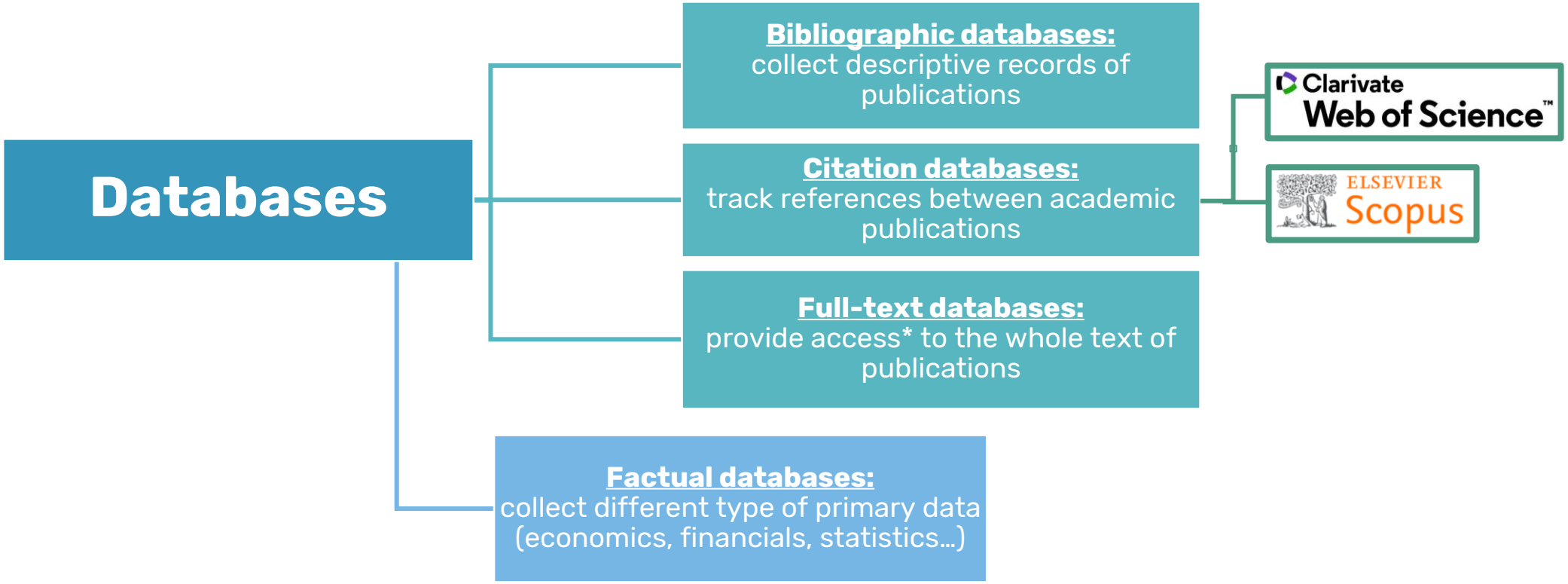
Transparency matters

always record how AI was used, verify its output, and never upload confidential or sensitive research data without authorisation

Human in-the-loop

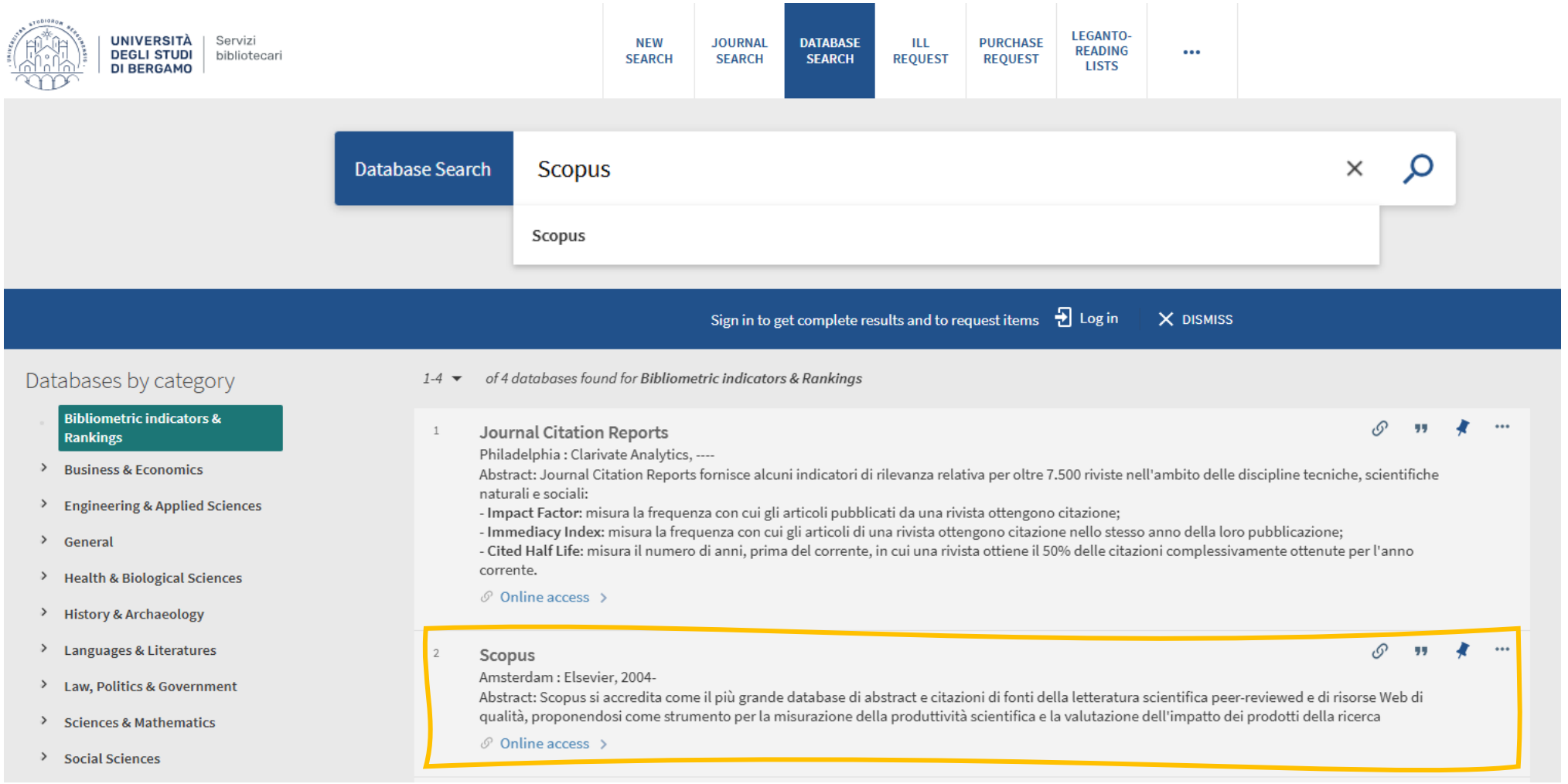
- **Define the question** and check the records
- **Set** inclusion and exclusion **criteria**
- **Decide eligibility** and resolve disagreements
- Verify quality and **risk of bias**
- **Verify every citation**, number and PRISMA count

Academic databases



*Note: access to the full text of the publication depends on the subscriptions held by the university

Using *Database search*



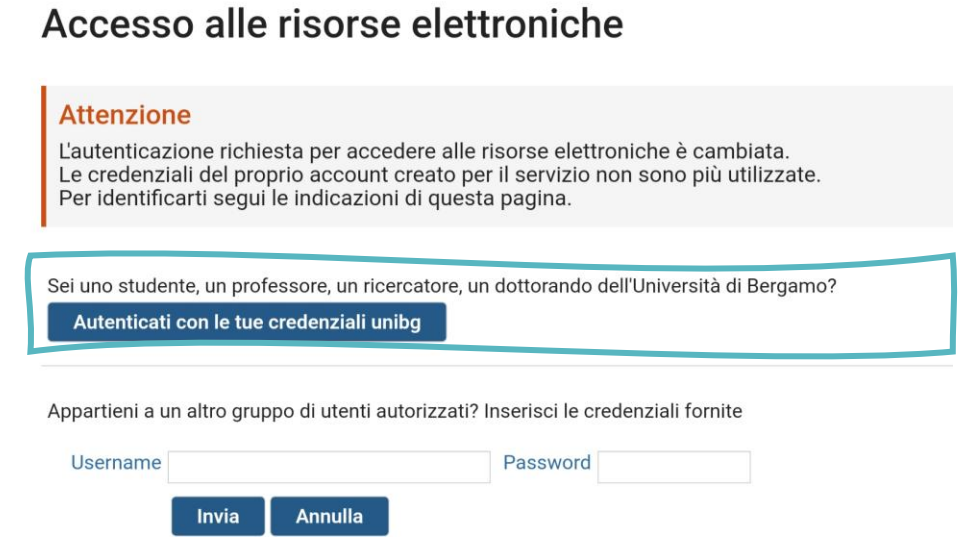
In campus and off campus access



Always access all resources **starting from the library services page**!
This allow you to access resources as **authorized user** even **outside the university network**

In campus: NO autentication required (some exceptions)

Off campus: you will be **asked to authenticate** using your institutional credentials to access the database or the full text of the publication



Focus on: Scopus

WHAT IS IT?

- Largest multidisciplinary database of peer-reviewed literature

COVERAGE

- Journals, books, and conference proceedings
  NOTE: not all publications are indexed in Scopus

FIELDS

- Science, Technology, Medicine, Economy, Arts & Humanities

RELEVANT FEATURES

- Advanced search tools, citation tracking, research analysis, and alerts

UPDATE FREQUENCY

- Daily with new publications

Take home message



Start with a well-defined **research question** and keep your search **focused**

Use multiple databases to get comprehensive coverage of relevant literature

Use **specific keywords** and combine them with Boolean operators to refine results

Evaluate abstracts before accessing full-texts

Be mindful of **publication dates, author and journal reputation**

Regularly **review and update** your search strategy

Use **PRISMA** for tracking your literary review screening

Don't hesitate **to ask librarian** for help

Library services

Finding resources	• use **BigSearch** to explore all the resources you can access through your library
Interlibrary Services	•get books (**ILL – Interlibrary Loan**) or chapters, articles, conference proceedings (**DD – Document Delivery**) not held by the Library
Reference service	• book an **appointment with the librarian** if you need help with bibliographic research or using the databases
Moodle	• basic skills to performing **bibliographic research**, avoid **plagiarism** and create your **references and bibliography**



KEEP CALM AND ASK YOUR LIBRARIAN

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